

Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based

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METHODOLOGY

1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

Parameter	Sentinel-1 C-SAR (IW Mode)
Frequency Band	C-band (~5.405 GHz)
Revisit Time	12 days (single satellite)
Spatial Resolution	5 m (range) × 20 m (azimuth)
Coverage Mode	Interferometric Wide Swath (IW)

2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

Multi-temporal **Sentinel-1 SLC IW mode** data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D
S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431
S1A_IW_SLC__1SDV_20250204T130238_20250204T130308_057747_071E86_3FE6
S1A_IW_SLC__1SDV_20250216T130238_20250216T130308_057922_072592_C312
S1A_IW_SLC__1SDV_20250228T130238_20250228T130307_058097_072CB4_7A14
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S1A_IW_SLC__1SDV_20250417T130238_20250417T130308_058797_0748F4_3FFD
S1A_IW_SLC__1SDV_20250429T130239_20250429T130309_058972_07501E_08D8
S1A_IW_SLC__1SDV_20250511T130238_20250511T130308_059147_0756BA_DE13
S1A_IW_SLC__1SDV_20250523T130238_20250523T130308_059322_075CCB_14B2
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S1A_IW_SLC__1SDV_20250628T130236_20250628T130306_059847_076EE3_6BFD

3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimetre-scale deformation over time:



VISUALIZATIONS

1. Defining specific bridge location & parameters for satellite SAR based inspection



Project & Bridge Details

State – Rajasthan
Bridge No. – 899/1
Bridge Chainage –905+300

Project - Chittorgarh to Kota

Zone – RJ 06

Structure Type – MNB

Sar Parameters

Sensor – Sentinel-1 (C-Band)
Beam Mode – IW (Interferometric Wide Swath)
Path – 34
Frame – 519

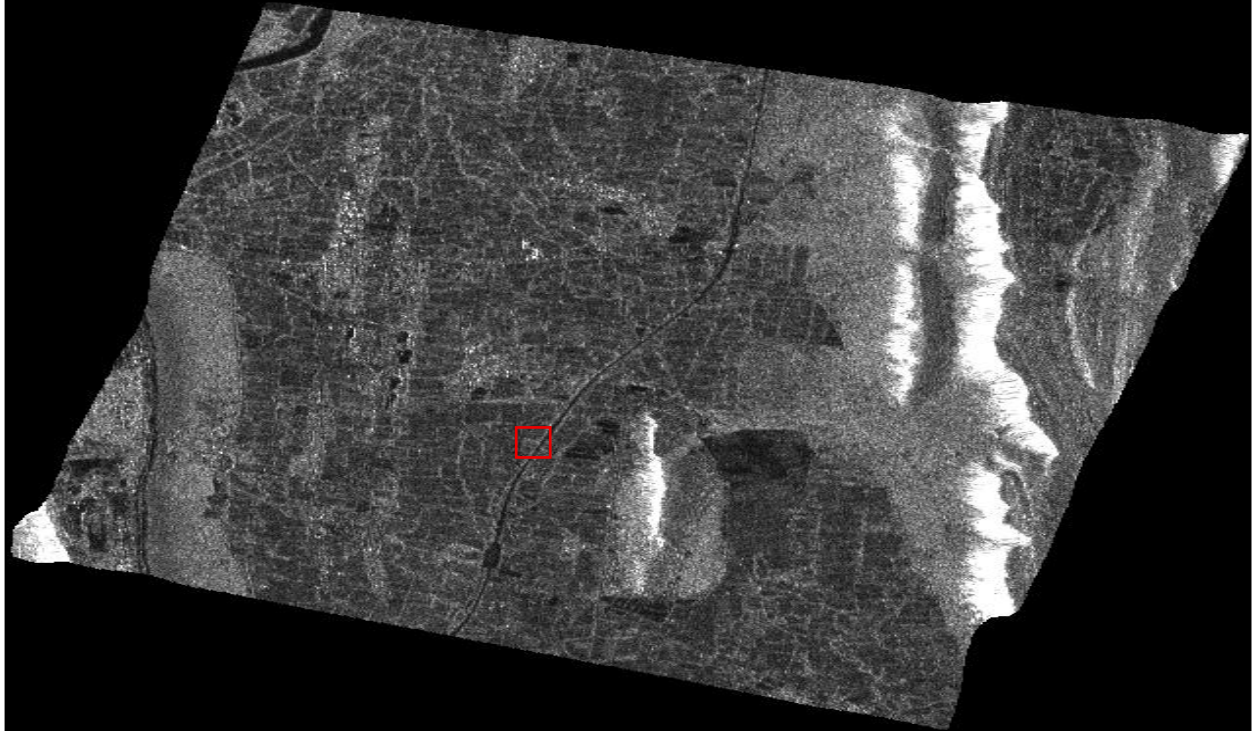
Flight Direction - DESCENDING

Polarization – VV+VH

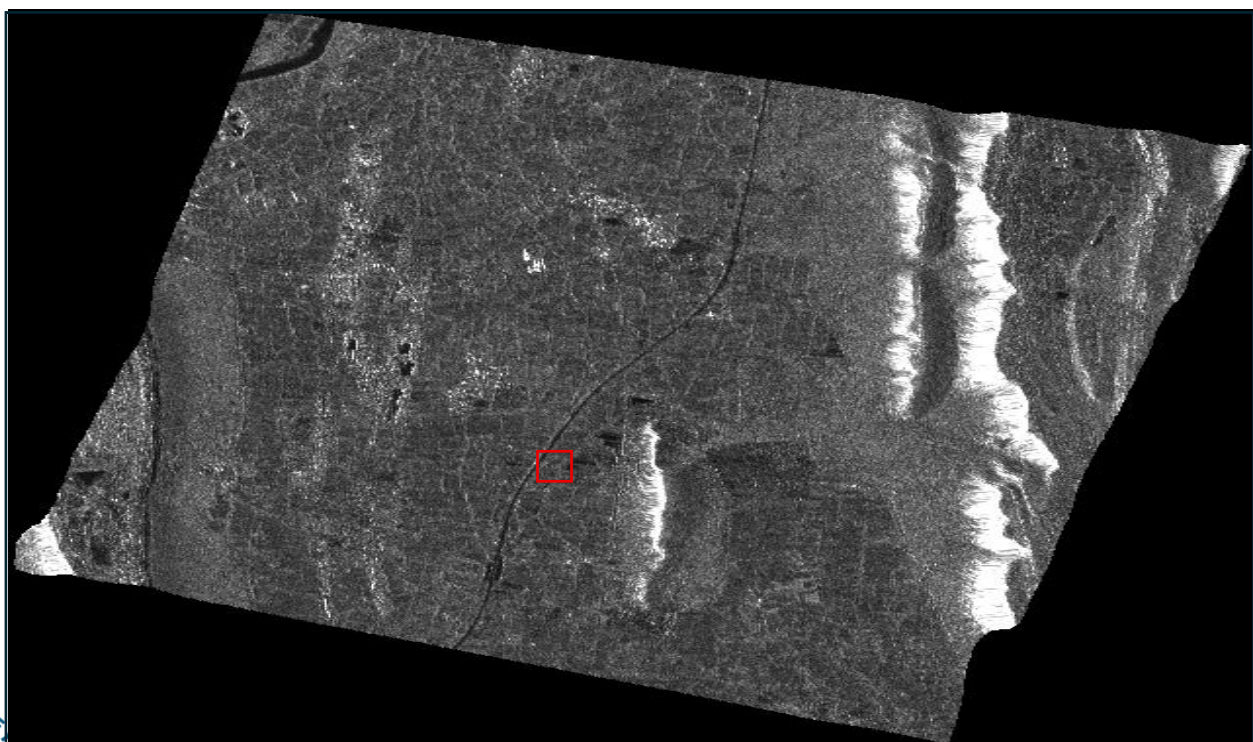
Absolute Orbit – 54706

2. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Orthorectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization

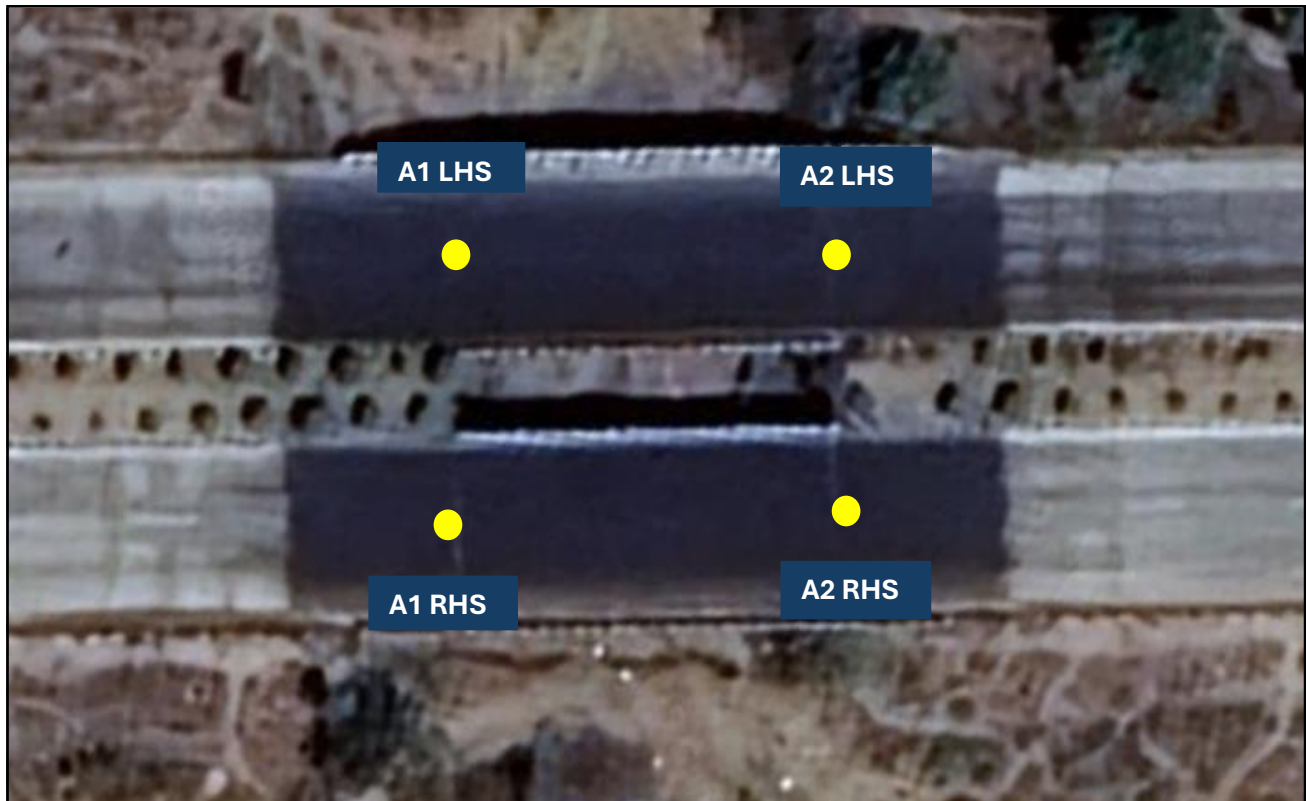


3. Results of SAR based bridge inspection, including visualizations

Bridge Components

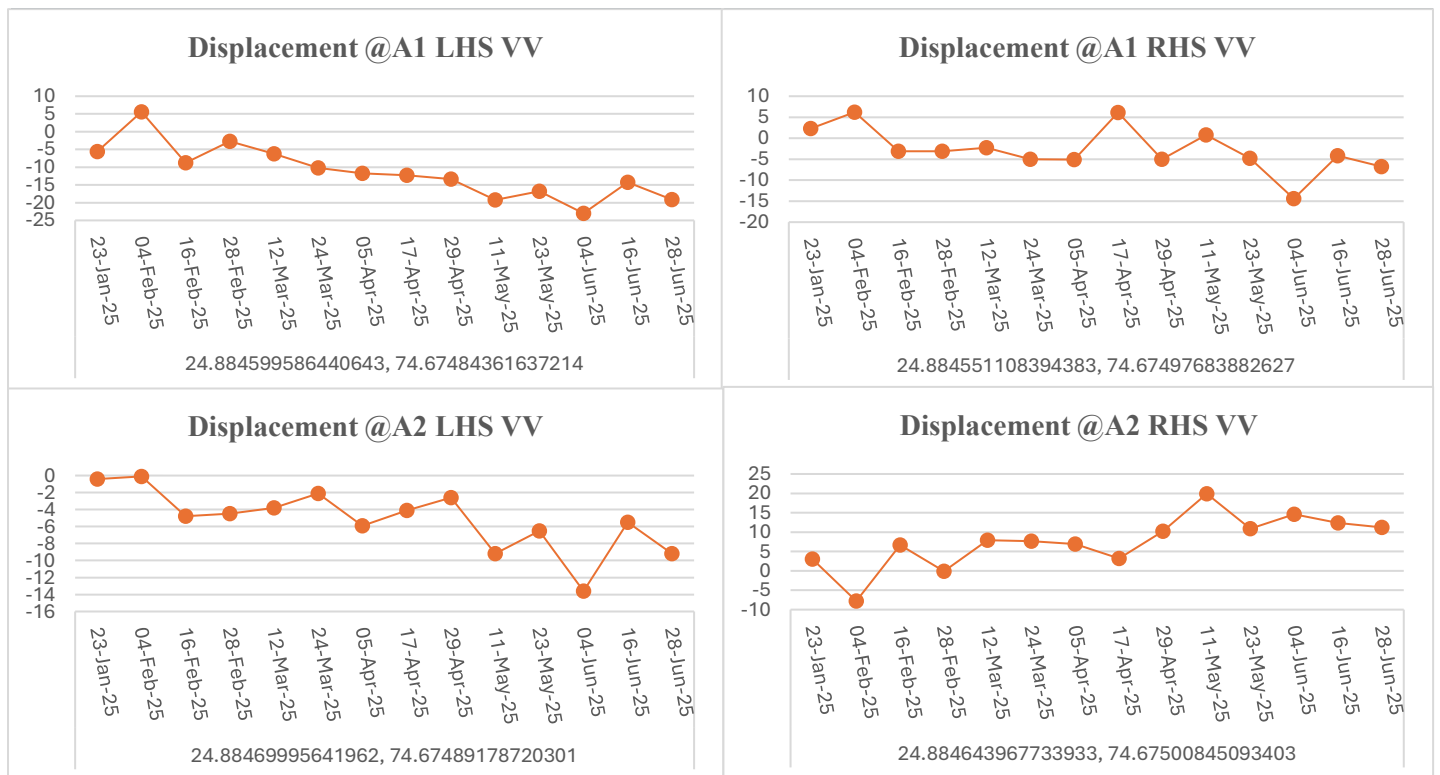
MNB – CH-905+300

Length – 21.37M

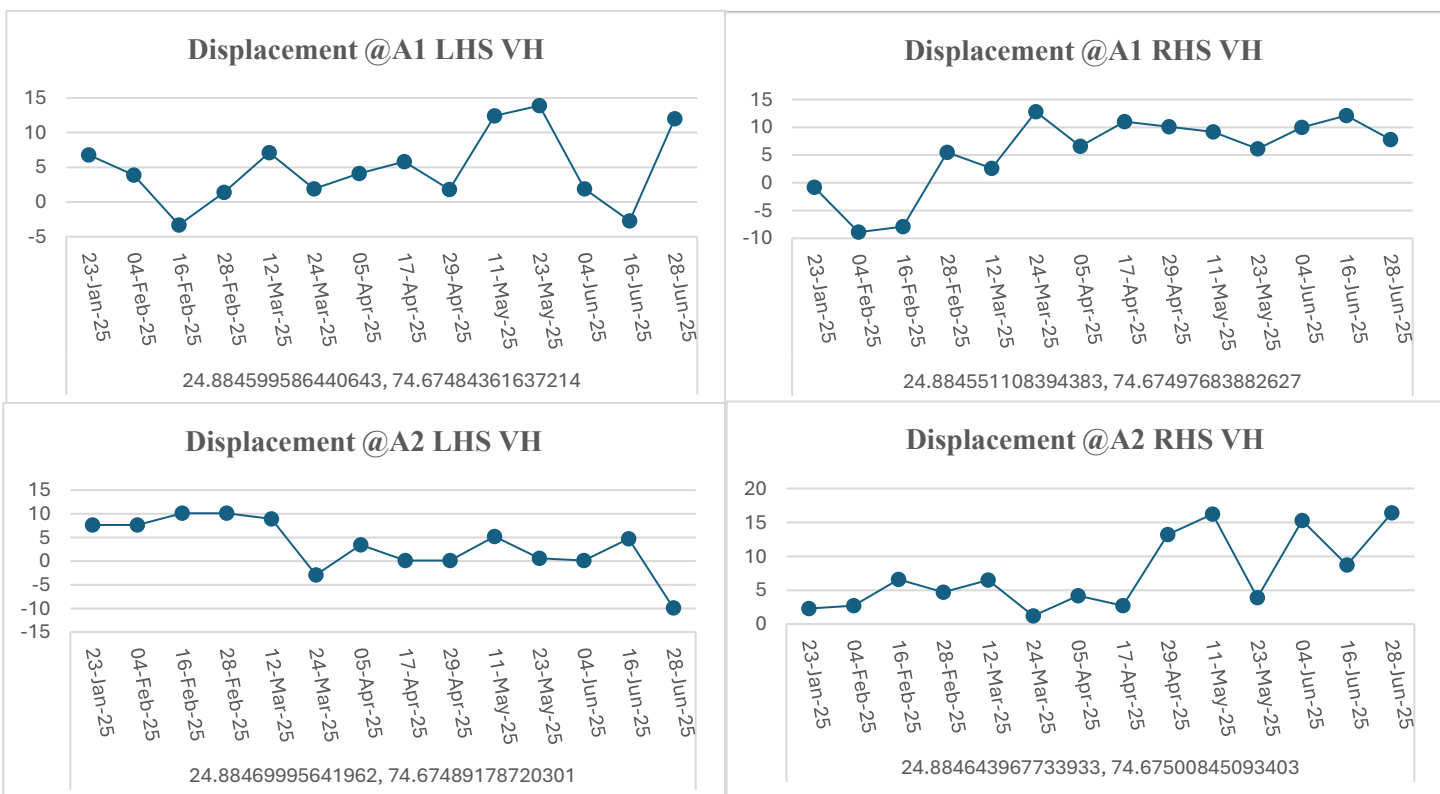


Scan Duration		From : 23 Jan 2025	To : 28 June 2025
Components & Abbreviation			
A1		Abutment 1	
A2		Abutment 2	
LHS		Left Hand Side	
RHS		Right Hand Side	

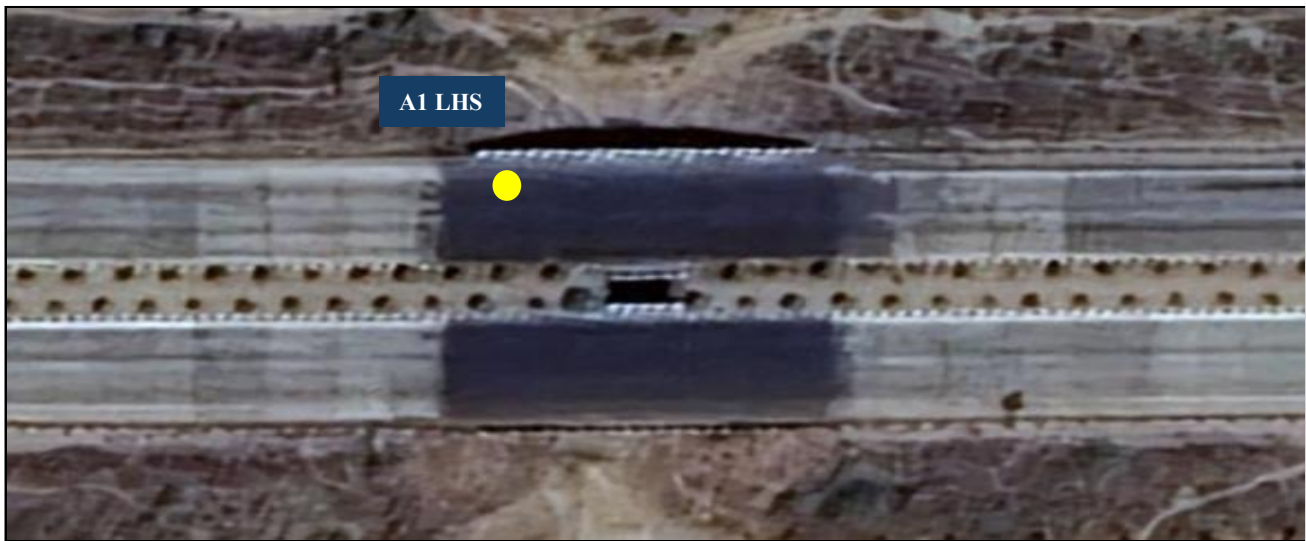
3.1 VV Displacement from 23 Jan 2025 To 28 June 2025



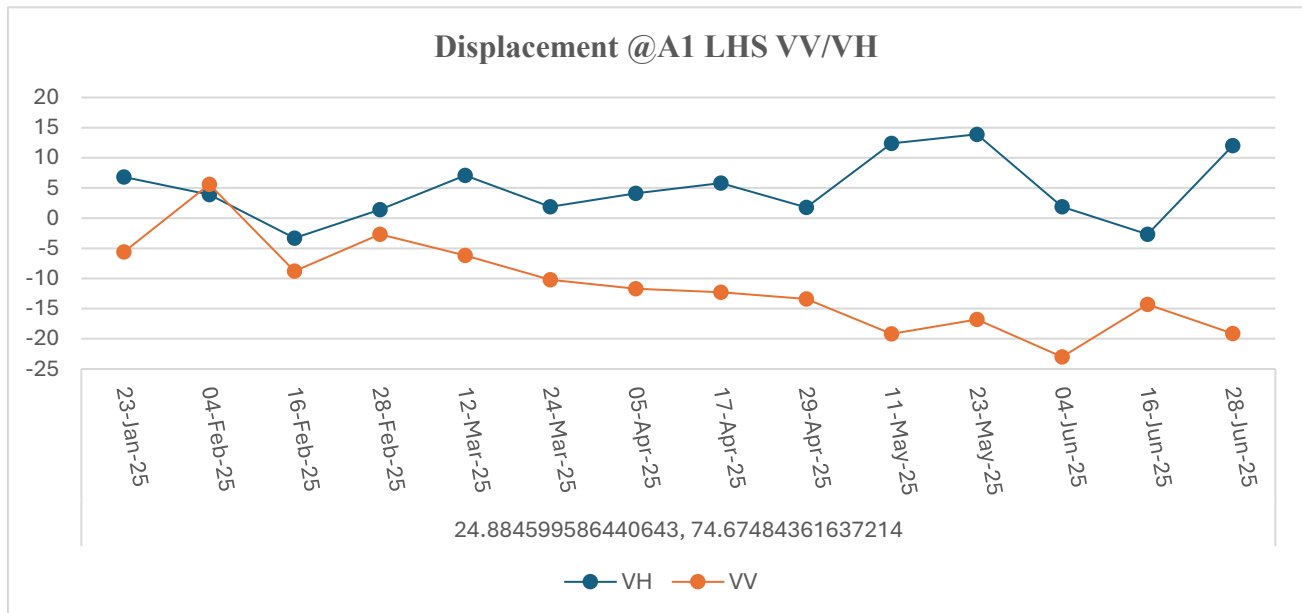
3.2 VH Displacement from 23 Jan 2025 To 28 June 2025



3.3 SAR Based Analysis at Abutment 1 LHS:



A1 LHS (24.884599586440643, 74.67484361637214)



Key Observations:

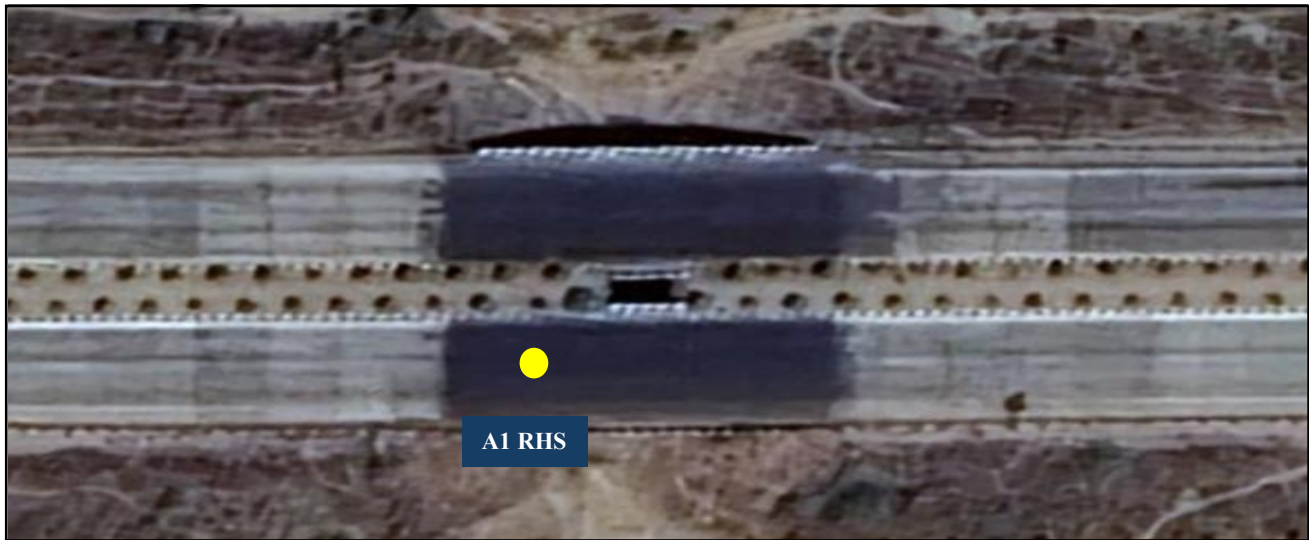
Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** VH values are predominantly positive, ranging approximately from +1 mm to +14 mm, indicating a net lateral movement in a consistent direction at A1 LHS.
- **Vertical Displacement (VV)** VV values are predominantly negative, ranging approximately from -5 mm to -24 mm, indicating net downward vertical movement (settlement).

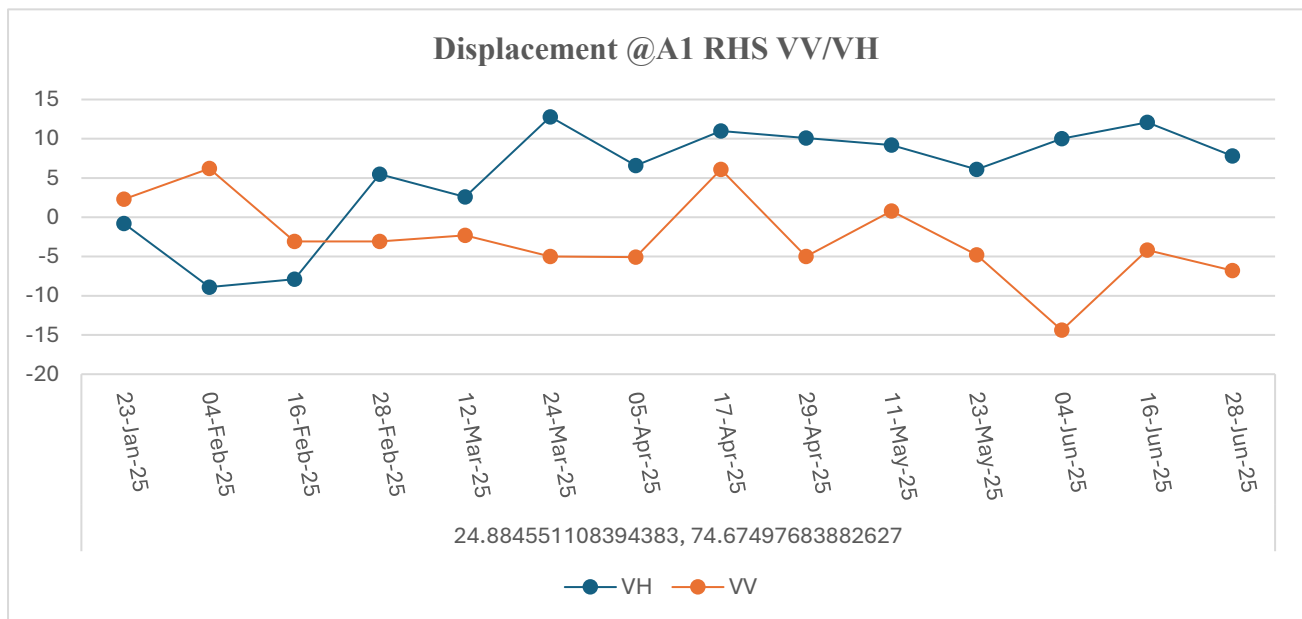
Critical Events & Anomalies:

- **February 2025:** Opposing movement directions emerge clearly, with VH remaining positive while VV turns increasingly negative, indicating combined lateral translation with vertical settlement.
- **June 2025:** A period of simultaneous VH increase and VV deepening suggests rotation or differential support behaviour at A1 LHS, potentially influenced by bearing response, soil–structure interaction, or thermal gradients.

3.4 SAR Based Analysis at Abutment 1 RHS:



A1 RHS (24.884551108394383, 74.67497683882627)



Key Observations:

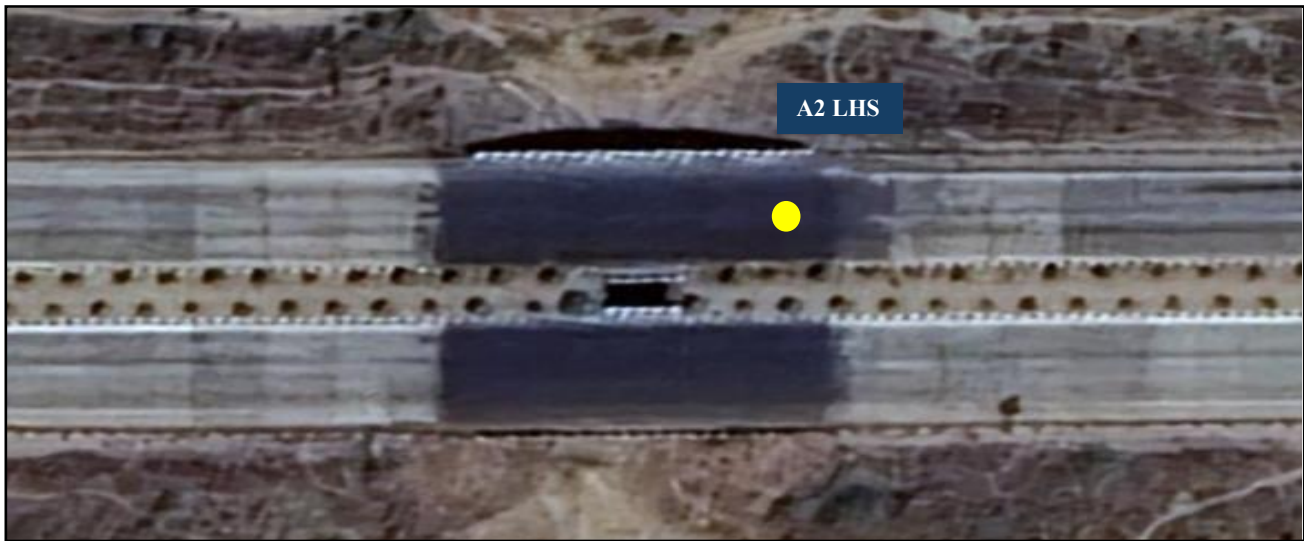
Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** values exhibit predominantly positive behaviour, generally ranging from approximately +3 mm to +13 mm, indicating a consistent lateral displacement trend at A1 RHS.
- **Vertical displacement (VV)** VV values are predominantly negative, ranging approximately from +6 mm to -15 mm, indicating net downward vertical movement (settlement).

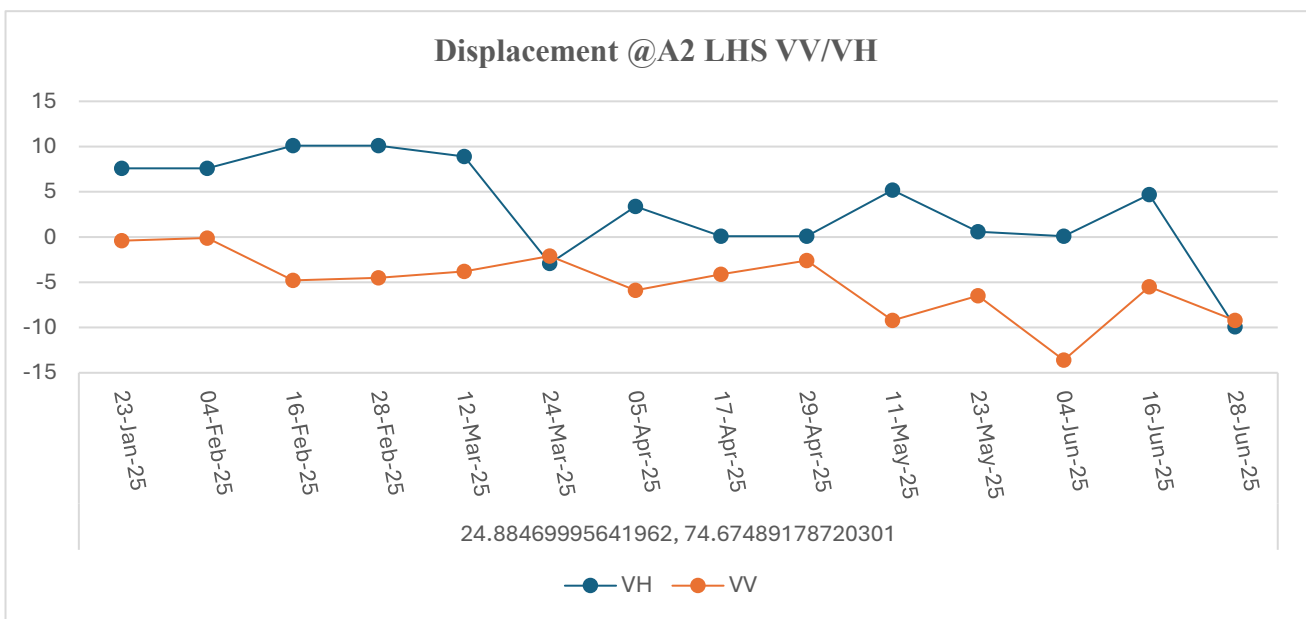
Critical Events & Anomalies:

- **February 2025:** A clear sign change in VH (from negative to positive) coincides with VV trending downward, suggesting structural realignment or bearing activation rather than measurement noise.
- **March-April 2025:** Simultaneous high VH values and increasing VV settlement indicate a combined translational-rotational response at the abutment level.
- **June 2025:** The pronounced negative VV excursion (-15 mm) with sustained positive VH suggests foundation compression or support stiffness variation, potentially influenced by seasonal or subsoil effects.

3.5 SAR Based Analysis at Abutment 2 LHS:



A2 LHS (24.88469995641962, 74.67489178720301)



Key Observations:

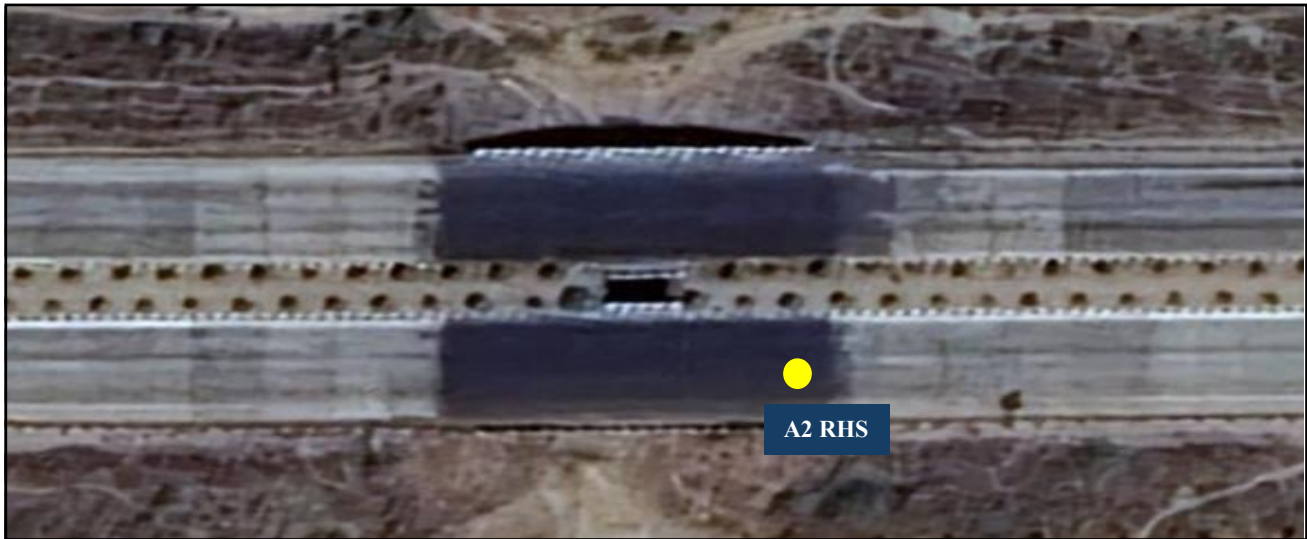
Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** values are predominantly positive throughout the monitoring period, generally ranging from approximately 0 mm to +10 mm, indicating stable lateral movement without reversal at A2 LHS.
- **Vertical Displacement (VV)** VV values are consistently negative, generally ranging from approximately -2 mm to -14 mm, indicating persistent downward vertical movement (settlement) at A2 LHS.

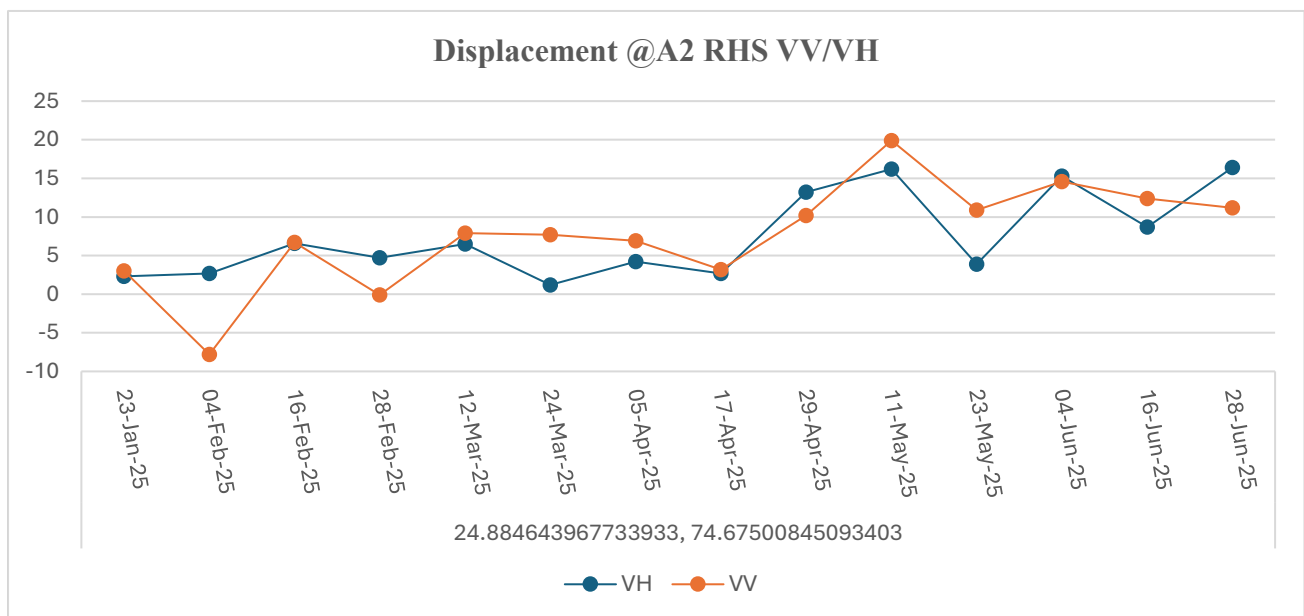
Critical Events & Anomalies:

- **March 2025:** A short-duration reduction in VH combined with increased VV settlement indicates a temporary stiffness change or load redistribution at the abutment or bearing level.
- **June 2025:** The sharp negative VV excursion in early June corresponds with only minor VH variation, suggesting vertical compression-dominated behaviour rather than combined translation.

3.6 SAR Based Analysis at Abutment 2 RHS:



A2 RHS (24.884643967733933, 74.67500845093403)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical (VV) displacement** values show greater variability compared to VH, ranging from approximately –8 mm to +20 mm.
- **Horizontal (VH) displacement** values are predominantly positive over the monitoring period, generally ranging from approximately +2 mm to +16 mm.

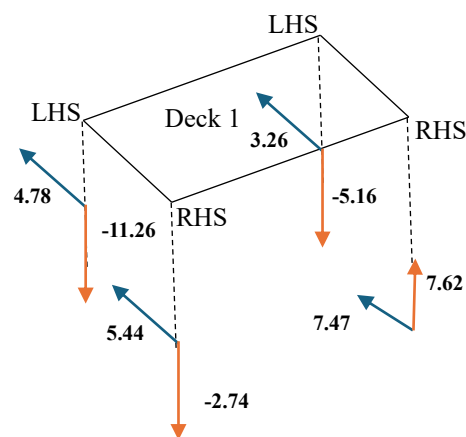
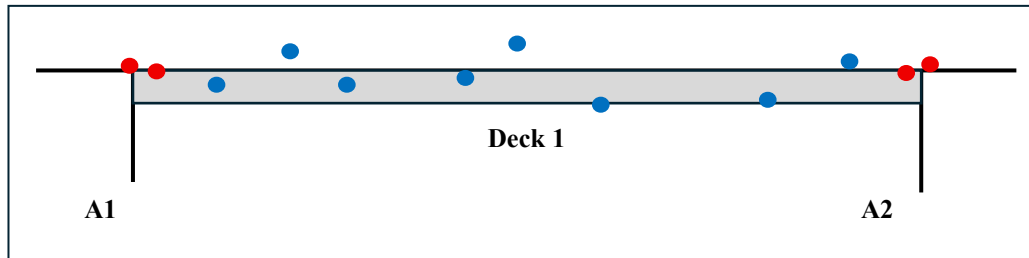
Critical Events & Anomalies:

- **February 2025:** A pronounced negative VV excursion occurs with relatively low VH response, indicating a short-term localized settlement event without significant lateral coupling.
- **May 2025:** A sharp drop in VH accompanied by reduced VV magnitude indicates a temporary horizontal stiffness change, likely reflecting restraint activation.
- **June 2025:** VV remains elevated while VH fluctuates within a moderate range, indicating vertical-dominant.

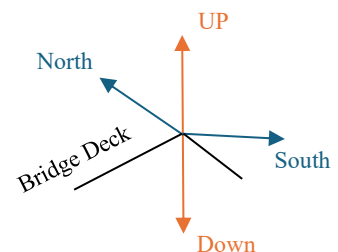
ANALYSIS

1. Identify potential deformation, shifts or movement of the bridge structure

- Points of Analysis (POA) of bridge (●)



Bridge Deck Slab Analysis



Vertical Displacement Behaviour:

- The bridge deck slab exhibits distinctly non-uniform vertical displacement, indicating pronounced differential movement between support locations. This opposing vertical response across the deck confirms the presence of rotational deformation, rather than uniform vertical translation.

Horizontal Displacement Behaviour:

- Horizontal displacement shows clear directional asymmetry across the deck slab, with varying magnitudes at each support. The largest lateral displacement is recorded at the RHS lower region, reaching approximately +7.47 mm, while the LHS regions show comparatively smaller but still significant movements in the range of 3.26 mm to 5.44 mm

CONCLUSION:

- The bridge deck slab demonstrates a non-symmetric deformation pattern, characterized by uplift at the RHS, settlement toward the LHS/central regions, and uneven horizontal displacement distribution. The coexistence of vertical uplift on one side and settlement on the opposite side, combined with differential horizontal movement, strongly indicates torsional rotation of the deck slab, likely influenced by bearing response, boundary conditions, or load eccentricity rather than uniform structural movement.